

LOST IN THE MES(h)S

A BATTLE-HARDENED MESH STACK / BUILT FROM THE METAL UP

Encrypted. Robust. Frugal.

Raw 802.11 injection gives the freedom and options. We built the solution.

ENCRYPTED

Forward-secret AEAD over every byte.

- ▶ ChaCha20-Poly1305 AEAD
- ▶ HKDF ratchet · 60s epoch · zeroize
- ▶ 128-bit per-sender replay window
- ▶ Padded to MAX_PLAINTEXT — LPI/LPD

ROBUST

Self-healing under jam + partition.

- ▶ Atomic channel hop at epoch boundary
- ▶ PRR splits jammed vs dead — no false evict
- ▶ Counter-suppressed flood — partition heals
- ▶ Restart-tolerant replay window

FRUGAL

Minimum bytes for maximum reach.

- ▶ High-MCS unicast — 5–30 Mbps
- ▶ RaptorQ adaptive rate — exact symbols
- ▶ Coverage-gated termination — bitmap ACKs
- ▶ 1× flood overhead at any density

Five crates, one vertical slice.

L4 · APPLICATION
app_sdk + api

NODE API · REST · WS · SVELTE

L3 · MESH
mesh

BEACONS · FLOOD · DIJKSTRA · CH-HOP

L2 · DELIVERY
delivery

RAPTORQ · ADAPTIVE RATE · COVERAGE

L1 · TRANSMISSION
transport + radio

CHACHA20-POLY1305 · 802.11 INJECT

L0 · SHARED
CONTRACT

common

NodeId · ObjectId ·
Epoch

PacketMeta ·
NeighborInfo

ObjectBitmap

ARC < DYN TRAIT >

Every layer talks to its
neighbors through a trait.
Every test mocks that trait.

Every byte on the air.

Raw 802.11 unicast to a sentinel MAC. High-MCS rates — never the broadcast fallback.



AEAD, ratchet, replay window.

KEY FLOW · PER PACKET

• **K_ROOT** provisioned 00B · zeroize on drop

↓ HKDF-SHA256(epoch, "kova-mesh/ratchet/v1")

• **K_EPOCH** window { e-2, e-1, e, e+1 }

↓ ChaCha20-Poly1305.seal

NONCE 12B

sender || counter

AAD 22B

full header

PLAINTEXT

kind || len || pad

• **CIPHERTEXT + 16B TAG** → radio

★ FORWARD SECRECY

HKDF epoch ratchet · past keys zeroized on advance.

★ LPI / LPD

Kind byte inside ciphertext · length-indistinguishable.

★ REPLAY GUARD

Per-sender 128-bit window · checked *after* AEAD.

★ DOS-BOUNDED RESYNC

Resync ≤ 5 epochs ahead · past cap → AuthFailed.

Fountain code with a coverage gate.

One broadcast stream serves every receiver. RaptorQ keeps generating symbols; the sender stops when peer beacons confirm decode.



Passive neighbor discovery.

No unicast probes. No routing convergence. Every node reconstructs the full topology from what it overhears.

BEACON

100 ms

±10% jitter

```
Carries: epoch, seq, neighbors_heard[],
decoded bitmap.
```

- EVICT @ 2 000 MS

PRR ESTIMATION

Blend signal strength with actual delivery — take whichever looks worse.

R S S I how loud the radio reads

DELIVERY how many beacons actually landed

SMOOTH EMA over time — no flapping

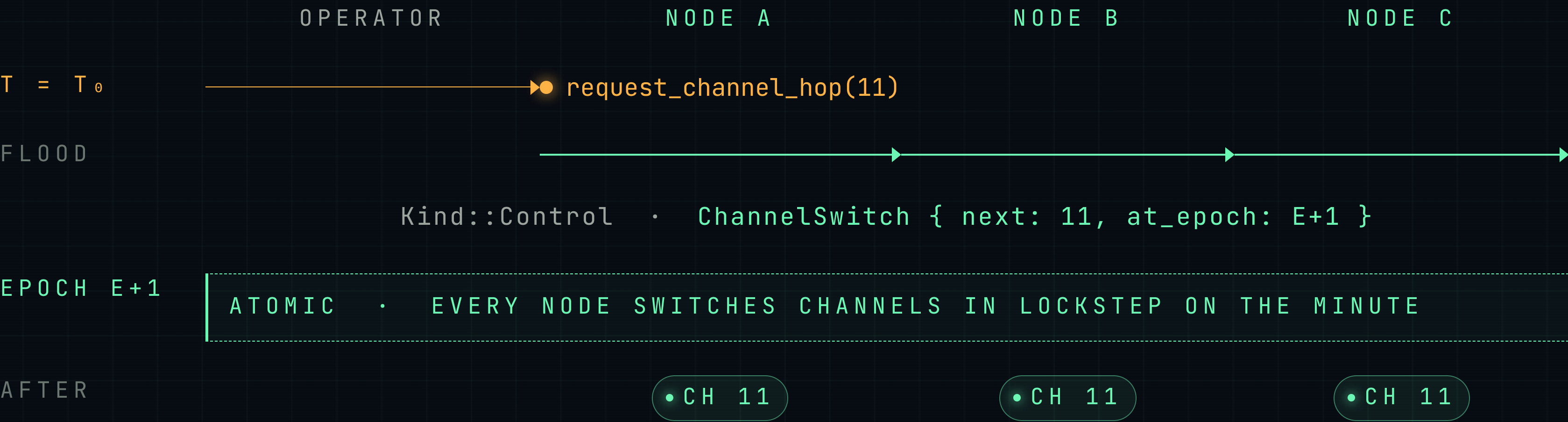
• CONSERVATIVE BLEND

COUNTER-SUPPRESSED FLOOD

- ① Receive new packet
- ② Delay $\text{rand}(0 \dots 50 \text{ ms})$
- ③ Count overheard copies
- ④ If $\text{count} < 2 \rightarrow \text{forward}$
- ✗ else cancel — region self-quiets

- $1 \times \text{OVERHEAD} \cdot \text{ANY DENSITY}$

Atomic channel hop. Zero coordination overhead.



SDK, API, dashboard.

APP_SDK · ONE-LINE NODE

```
// build & start
let node = NodeBuilder::new(id, "pass", "wlan0").build()?;

// reliable delivery
node.send(Text { content: "grid 472".into() })?;
node.send(Image { mime, bytes })?;

// unreliable low-latency video
stream.send_frame(&rgb, w, h)?;

// mesh awareness
let peers      = node.neighbors();
let topology   = node.topology();
node.request_channel_hop(11)?;
```

API · REST + WS / AXUM

```
GET  /api/data      state · neighbors · topology
POST /api/send      text · base64 image
POST /api/video/frame      unreliable lane
GET  /api/video/stream/:id  MJPEG multipart
WS   /api/raptor/ws      live FEC telemetry
POST /api/channel/hop      coordinated switch
```

DASHBOARD · SVELTE

Live topology map · PRR-weighted edges · RaptorQ matrix
animation · remote MJPEG feeds · operator channel control.

What it survives.

THREAT

MITIGATION

Passive eavesdrop

ChaCha20-Poly1305 AEAD; Kind byte inside ciphertext.

Replay attack

Per-sender 128-bit sliding window *after* AEAD.

Node capture / key extract

HKDF ratchet. Past keys zeroized. Forward secrecy holds.

Network partition

Counter-suppressed flood recovers on heal.

THREAT

MITIGATION

Single-channel jammer

Coordinated channel hop at epoch boundary. Atomic.

Spoofed frames

AEAD tag rejection. Wrong key → AuthFailed.

Jammed link vs. dead peer

PRR distinguishes. Silence evicts; PRR collapse does not.

Node restart

Replay window detects counter reset. Peer re-accepted.

Where the savings come from.

UNICAST VS BROADCAST

5–30 Mbps

vs 1–6 Mbps broadcast fallback. High-MCS to a sentinel MAC.

BEACON FOOTPRINT

14×

~100 B natural vs ~1 400 B if padded.
Beacons stay tiny; data padded for LPI.

FLOOD OVERHEAD

1×

Any density. Dense regions self-quiet via counter suppression.

+ MECHANISMS

- ▶ RaptorQ adaptive rate — exactly enough symbols for observed PRR.
- ▶ Bitmap beacon piggyback — coverage ACKs cost zero extra packets.
- ▶ 2.4 GHz HT20 MCS 0 — best sensitivity per milliwatt.
- ▶ Coverage-driven termination — stops when all targets decode.
- ▶ Video lane fire-and-forget — no FEC overhead on latency traffic.
- ▶ FEC + Control padded to MAX_PLAINTEXT — LPI without bloating beacons.

Hours away. The primitives are done.

What we shipped is a platform. Each row below is a Custom message type or a callback away.

HOURS · READY TO BUILD	
Distributed target tracking	Custom + Kalman
Progressive ISR image relay	multi-path FEC
Swarm leader election	beacon priority
Spectrum-aware reroute	auto channel-hop

DAYS+ · ROADMAP	
Noise IK pairwise join	~1 day · snow
GPS / IMU telemetry fusion	~1 day
Full MLS group encryption	~2 days
kHz-rate FHSS	HW-limited